

All three frontends: RAM over a whole race

Track N Race · Qt, Electron and the minimal recorder · one full race each, live at 1x · 9 October 2026

Summary

- **At the end of the race (99 minutes of replay, 71 laps): Electron 714 MB, Qt 444 MB, the minimal recorder 84 MB.** Electron ends about 1.6× Qt.
- **Peaks: Electron 748 MB, Qt 463 MB, the recorder 85 MB.** No app ran away; each peak sits within about 40 MB of its own trend.
- **Both apps with charts grow steadily through the race:** Qt by about 103 MB an hour, Electron by about 80 MB an hour; the recorder by about 12. The largest single part is the shared engine's in-memory copy of the recording: 81 MB at the end of the race in Qt (measured).
- **The minimal recorder stays between about 50 and 85 MB for the whole race,** rising slowly from about 65 to 84 MB as its file index grows, and drops to 30 MB once the recording is closed.
- **Most of Electron's extra cost is there from the start:** about 220 MB above Qt at minute 2, about 270 MB at the end.

Windows 11, RTX 4050 laptop GPU, 1920 × 1080 at 125 %. Data: the 92-minute São Paulo race, as the UDP capture (`udp_capture.bin`, replayed with `play_udp.py`) or as its recording (`udp_capture_sao_paulo_v6_2026-10-07.tnrd`). **Private memory** is memory only this program uses; it is the number to compare. The **working set** is what currently sits in physical RAM, and Windows trims it freely. Each app ran the whole capture (99 minutes: the formation lap and 71 laps) on its own, one after another, fresh from launch, on the same machine. Qt and Electron showed Overview with Current Lap and nothing else was touched; Electron ran with React Scan off. All three were recording the race to a file, as they do by default. Each app's processes were read from Windows every 5 s.

1. A whole race, live

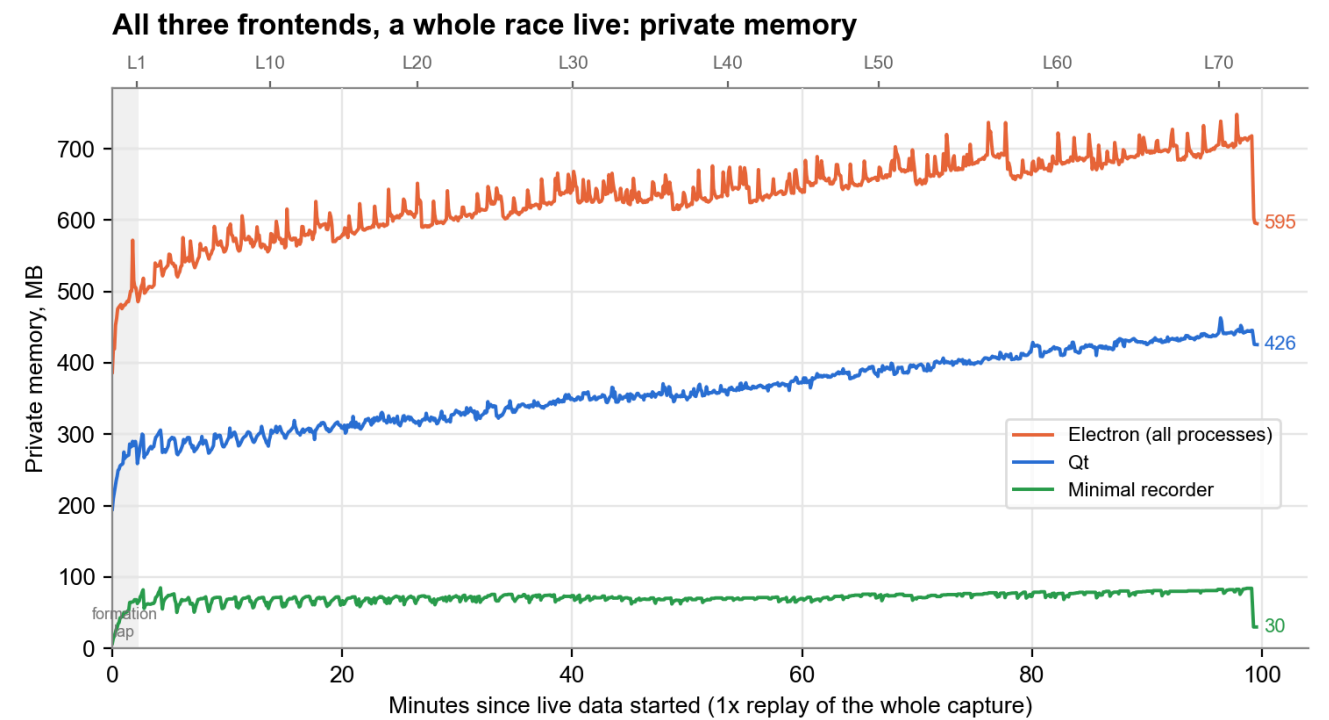


Figure 1. Private memory of each app over the whole race (Electron summed over its processes). Lap numbers run along the top; the shaded start is the formation lap. The number at the end of each line is its last sample, after the replay stopped.

	Electron	Qt	Minimal recorder
Minute 2	500	283	65
Minute 15	573	302	66
Minute 30	603	330	72
Minute 60	645	373	69
Minute 90	696	428	80
End of the race	714	444	84
Peak (minute)	748 (98)	463 (96)	85 (4)
Growth, last hour, MB per hour	80	103	12
Working set at the end	843	408	94

All figures in MB, medians over a minute around each point. Growth is a straight-line fit from minute 35 to the end.

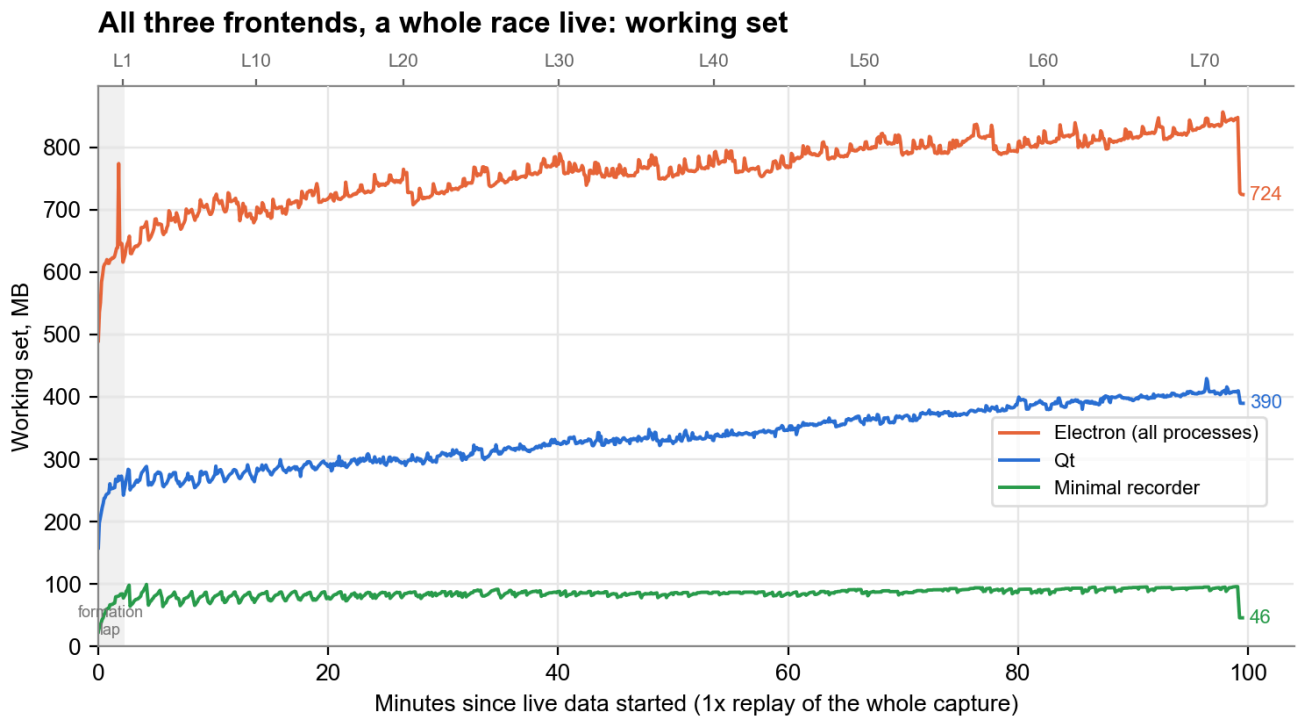


Figure 2. The same runs in working set: what each app had in physical RAM. Windows trims it freely, so it moves more than private memory; Electron's is higher because each of its processes keeps its own.

2. Inside each app

Electron

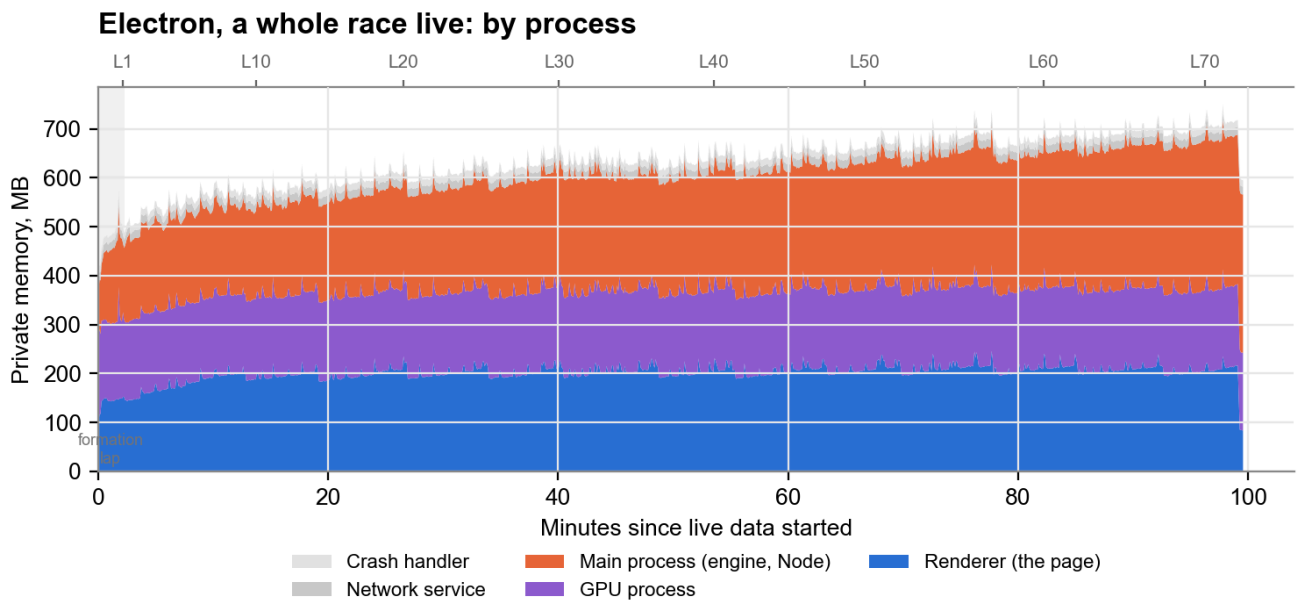


Figure 3. Electron by process over the race. The main process, which runs the engine, carries most of the growth (+140 of +214 MB); the renderer levels off after about 15 minutes and the GPU process stays flat.

	Total	Main	GPU	Renderer	Network	Crash handler
Minute 2	502	166	159	148	16	13
Minute 15	573	188	165	191	16	13
Minute 30	603	213	164	197	16	13
Minute 60	644	250	163	202	16	13
Minute 90	697	291	167	209	16	13
End of the race	714	306	164	214	16	13

Qt

Qt, a whole race live: what the private memory is

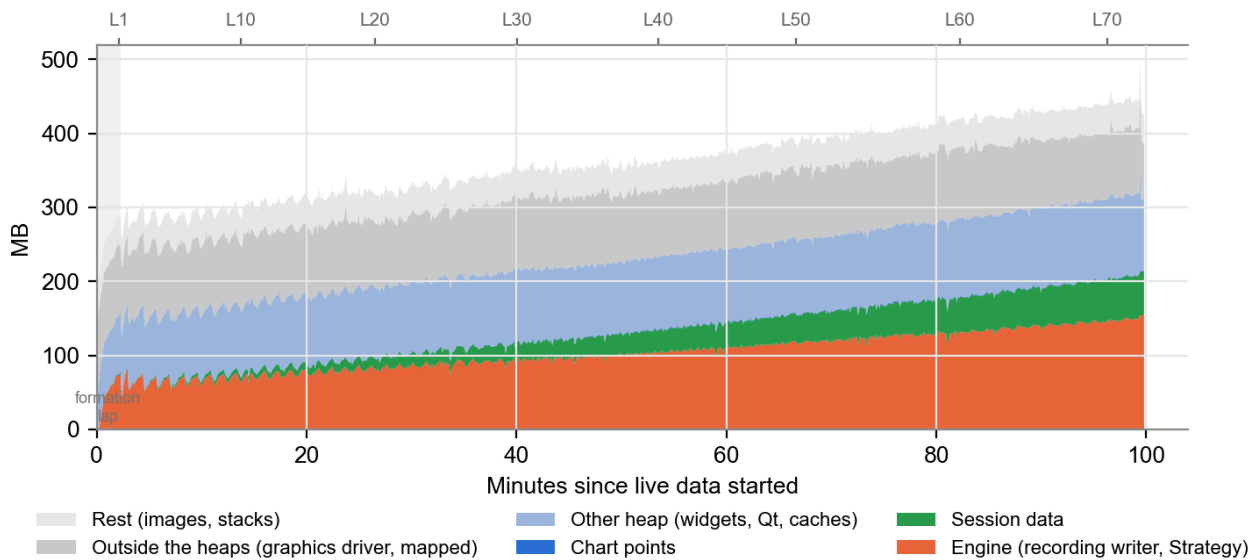


Figure 4. Qt's private memory split by owner, from its own memory log (one sample a second). The layers add up to the total.

Qt, a whole race live: the parts that grow with the race

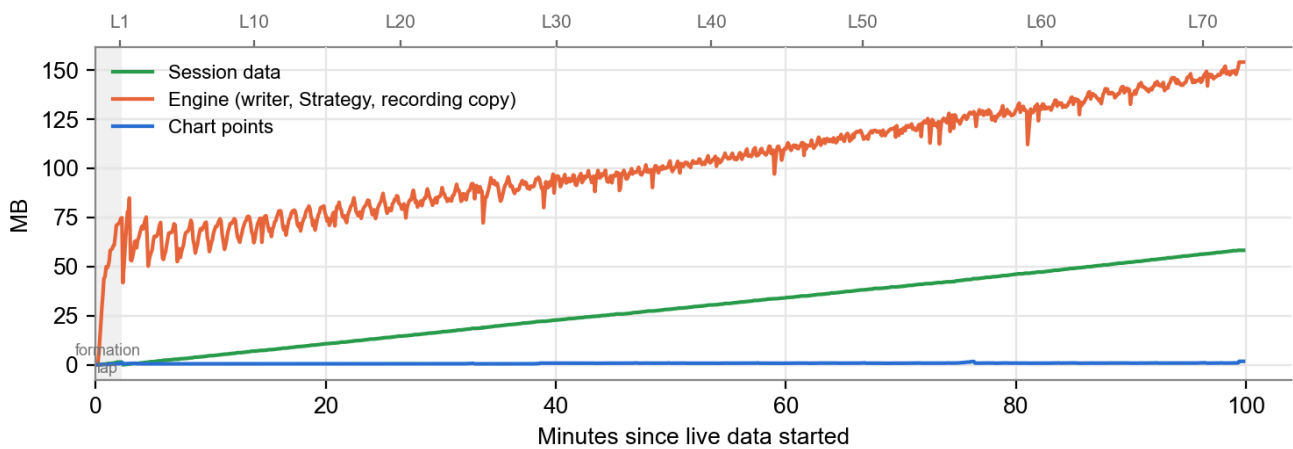


Figure 5. The parts of Qt that grow with the race. The engine's growth is mostly the in-memory copy of the recording; the session data is every sample of the race; chart points are only the page on screen.

	Private	Engine	Session data	Chart points	Other heap	Outside heaps
Minute 2	282	71	1	0.6	78	94
Minute 30	328	86	17	0.5	95	92
Minute 60	372	110	34	0.8	99	90
End of the race	445	150	58	1.0	109	88

Minimal recorder

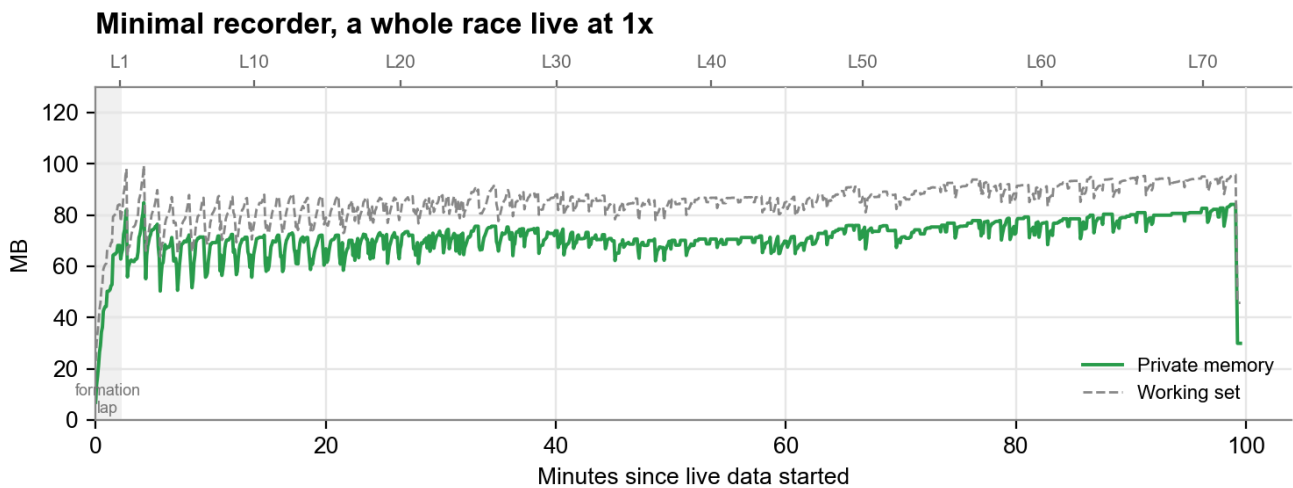


Figure 6. The minimal recorder over the race, at 1x.

3. Playback, from the earlier runs

The minimal recorder doesn't play recordings back, so playback isn't part of the full-race runs. From the reports' end-of-race All Laps tour (8 October), the same recording and pages in both apps:

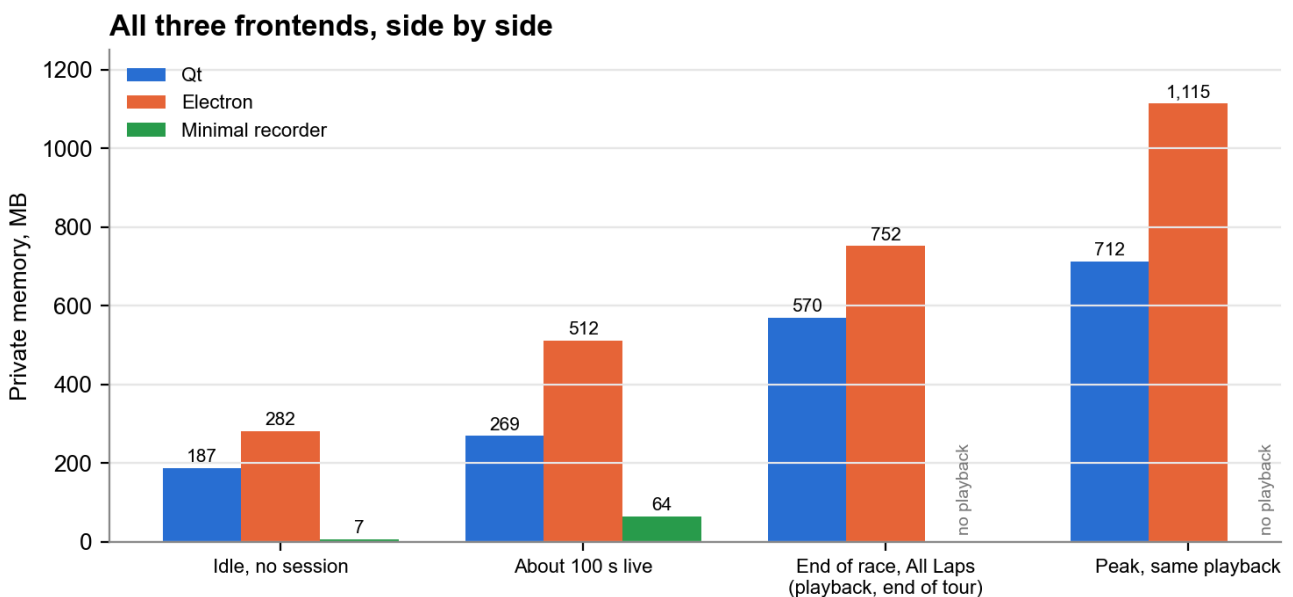


Figure 7. Earlier snapshots: idle, about 100 s live, and the end-of-race All Laps playback tour.

4. How comparable this is

- **Same data, same conditions.** Every app received the same 99-minute replay at real speed, from launch, alone on the same machine, with the same page and chart window, and wrote the same recording.
- **One run each.** Run-to-run variation measured earlier was about ± 10 MB on short runs; over a whole race the trends are far larger than that.
- **Overview only.** No page tour: opening other pages adds memory, as the per-app reports show.
- **Current Lap.** All Laps would add the whole-race chart points to Qt and Electron alike.

5. Observations

Over a whole race, Qt costs about 62 % of Electron, and the recorder a small fraction of either. None of the three leaks; all growth is data the race produces.

- **The growth is the engine and the live history.** In Qt the engine went from 71 to 150 MB, the session data from 1 to 58 MB and the rest of the heap by about 30 MB. In Electron the main process grew by 140 MB and the renderer by 66 MB. The largest single item is the engine's in-memory copy of the recording: at the end of the race Qt's engine held 70 MB of recording chunks and 11.5 MB of shared records that were also in the file. The reports' step 1 would move them to disk, taking about 80 MB off the end of a race in both apps.
- **Session data is modest.** Qt holds every sample of the race in about 60 MB at the end.
- **Electron's gap is mostly fixed cost.** Its renderer and GPU process settle at 360–380 MB together after about 15 minutes and stay there. Qt's side apart from the engine and session data stays at 210–240 MB.
- **The recorder is the floor.** It records the same race with the same engine, without the in-memory copy or any charts, and grows by only about 12 MB an hour (its file index). Everything Qt and Electron hold above it is their UI and the live history they keep for the charts.